

Agent 2 & Agent 3 Technical Architecture Documentation

Complete Specification of Claim Extraction, Web Search Verification, and Deception Detection

Project: ETRAI Fact-Checking Platform	Pipeline Stage: Agent 2 & Agent 3
AI Engine: OpenAI GPT-4o (JSON Mode)	Search Provider: Serper Search API
Source Files: claimExtractor.js & factVerifier.js	Status: Active Production

1. Executive Overview & Pipeline Context

ETRAI processes documents through a sequential 4-agent verification pipeline. While Agent 1 handles input parsing, **Agent 2 (Claim Extractor)** and **Agent 3 (Fact Verification Agent)** form the core analytical core of the platform:

- Agent 2 (Claim Extractor):** Isolates up to 25 verifiable factual claims from unstructured document text, filtering out subjective commentary.
- Agent 3 (Fact Verification Agent):** Queries live web search indexes, cross-references evidence via GPT-4o, validates live HTTP URLs to eliminate broken links, and detects major event fabrications.

2. Agent 2: Claim Extractor Service (claimExtractor.js)

2.1 Input Guardrails: Accepts extracted document text from Agent 1 and truncates it to the first **15,000 characters** (~3,750 tokens) to optimize API speed and context window allocation.

2.2 Core GPT-4o Strategy: Uses gpt-4o (temperature: 0.2) with strict JSON mode formatting (response_format: { type: 'json_object' }). Focuses on specific quantitative metrics, dates, financial figures, and direct assertions.

Agent 2 Extraction Prompt:

'You are Agent 2 (Claim Extractor) in an AI Fact-Checking system. Extract up to 25 of the most important, specific, and verifiable factual claims. Focus on quantitative metrics, financial figures, and statements verifiable by web search. STRICT RULE: Max 25 claims. No subjective commentary.'

2.3 Output JSON Contract (Capped at 25 Claims):

```
[
  {
    "id": "claim_1",
    "text": "Global cloud computing expenditure grew by 24% in Q3 2024.",
    "category": "Statistical Metric",
    "importanceScore": 95
  }
]
```

2.4 Algorithmic Heuristic Fallback (extractMockClaims): If OpenAI is unconfigured or encounters an API error, Agent 2 falls back to regex sentence splitting `/(?<=[?!])\s+/` and filters sentences containing factual key terms (e.g. *percent, dollar, company, report, million, billion*), returning up to 25 scored claims algorithmically.

3. Agent 3: Fact Verification Agent Service (factVerifier.js)

3.1 Web Search Querying (Serper API): Agent 3 queries Google Serper API for each claim, fetching up to 5 organic search items (title, URL, snippet, domain). Filters against trusted domains like *reuters.com*, *apnews.com*, *bbc.com*, *bloomberg.com*, *wsj.com*, *factcheck.org*, *.gov*, *.edu*.

3.2 Classification & Verdict Rubric: Evaluates claims against search snippets using GPT-4o (temperature: 0.0):

Verdict	Classification Logic	Confidence
Verified	Search snippets directly and explicitly corroborate the claim. Link must pass live HTTP check.	85% - 98%
False	Snippets directly refute claim OR claim asserts a major public event with 0 search record.	90% - 98%
Suspicious	Claim is ambiguous, minor uncorroborated statement, or link fails HTTP validation.	50% - 70%

3.3 Major Event Fabrication Detection (isMajorEventAssertion): When a claim describes major public leader actions, wars, or treaties (detected via regex: */prime minister|president|modi|biden|putin|declared war|invaded|nuclear test/i*) but search returns 0 results, Agent 3 marks it **FALSE** rather than ambiguous with 94% confidence.

3.4 Live Source HTTP URL Validator (validateSourceUrl): Executes an HTTP GET request with a 4,000ms timeout on selected source URLs. Drops dead links (404s). If all sources for a 'Verified' claim fail, the verdict is automatically downgraded to Suspicious or False.

3.5 Agent 3 Output Data Contract:

```
[
  {
    "claimId": "claim_1",
    "claimText": "Global cloud computing expenditure grew by 24% in Q3 2024.",
    "category": "Statistical Metric",
    "status": "Verified",
    "confidence": 92,
    "explanation": "Confirmed by BBC News – Global Technology Report...",
    "sources": [{ "title": "BBC News", "url": "https://www.bbc.com/news", "domain": "bbc.com" }]
  }
]
```

4. Summary Comparison Matrix

Feature	Agent 2: Claim Extractor	Agent 3: Fact Verification Agent
Primary Goal	Extract testable factual claims	Verify claim truthfulness via web search
Input	Raw extracted text (~15k chars)	Array of extracted claim objects
AI Model & Temp	gpt-4o (Temperature: 0.2)	gpt-4o (Temperature: 0.0)
External APIs	OpenAI API	OpenAI API + Serper Google Search API
Capping Rule	Max 25 Claims	Processes all claims from Agent 2
Safeguard	Filters opinion & subjective text	isMajorEventAssertion & HTTP URL check